

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Full Study Title: A Study to Learn About How Well Yearly Updates to the COVID-19 Vaccine Work to Protect People From COVID-19 and How Much Money People Spend on Healthcare for COVID-19

Short Title/Acronym: BNT162b2 Post-Marketing Surveillance Study

Protocol Number: C4591074

NCT Number: NCT06923137 **Sponsor Name:** Pfizer

Investigational Product: BNT162b2 (Tozinameran / COVID-19 Vaccine updated annual formulations)

Phase of Development: Observational / Phase 4 (Post-marketing Surveillance)

Medical Monitor: Pfizer Clinical Operations

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the real-world vaccine effectiveness (VE) of the yearly updated BNT162b2 COVID-19 vaccine formulations against COVID-19-associated hospitalizations, emergency department/urgent care visits, and outpatient encounters. **Secondary Objectives:** To estimate the direct healthcare expenditures, payer costs, and the overall economic burden associated with COVID-19 in vaccinated versus unvaccinated person-time.

2.2 Background & Rationale To address the continuous evolution of SARS-CoV-2 and provide real-world evidence on the protective benefits of the 2024-2025 updated formulations of the BNT162b2 vaccine. The study aims to understand the level of protection against severe outcomes among children and adults and the resultant economic impact of COVID-19 in the United States by utilizing nationwide administrative claims linked with state vaccine registries.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (Retrospective Cohort Study) **Control Type:** Unvaccinated/Unexposed person-time (evaluated as a time-varying exposure) **Blinding:** Open-label / Non-interventional **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: Data-dependent (Millions of individuals available in HealthVerity claims database; e.g., >6.9 million individuals analyzed in mid-season cuts) **Number of Sites:** Nationwide US claims database (focusing on states with mandatory COVID-19 vaccine reporting like California)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 6 months (assessed in adult ≥ 18 years and pediatric 6 months–17 years cohorts). 2. Continuous medical and pharmacy enrollment in a health insurance plan providing data to HealthVerity prior to the index date (≥ 12 months for adults, ≥ 6 months for children). 3. Continuous residency in a US state with mandatory COVID-19 vaccination reporting that contributes the entire state registry to HealthVerity.

4.2 Exclusion Criteria 1. Mismatches in sex and/or year of birth between any of the available datasets. 2. Individuals failing to meet the continuous database enrollment timeline limits (e.g., gaps of > 30 days in coverage). 3. (For specific analytical cohorts) Immunocompromised status or prior receipt of non-BNT162b2 vaccines when isolating BNT162b2 effectiveness.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Receipt of the updated BNT162b2 COVID-19 vaccine formulation (e.g., 2024-2025 dose). Vaccination status is modeled as a time-varying exposure, where "exposed person-time" begins 14 days post-vaccination. **Route of Administration:** Intramuscular injection (retrospectively tracked via claims/registries). **Duration of Treatment:** Single or multiple doses; observational follow-up spanning the defined database study period.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care routines, concomitant seasonal vaccines (e.g., influenza vaccine co-administration tracking). **Prohibited:** N/A (This is a non-interventional observational study).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Look-back Period	6-12 months pre-index
Establish patient characteristics, clinical history, risk factors, and baseline healthcare utilization. Index Date			
Day 0	Start of study period / Date of vaccination.		
Unexposed Time Days 0 to 13 Evaluated as unexposed or unvaccinated person-time. Exposed Follow-up Days 14+			
Track inpatient, outpatient, ED/UC encounters, and pharmacy claims via HealthVerity data.			

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Vaccine Effectiveness (VE) against COVID-19-associated hospitalizations and severe outcomes. **Measurement Method:** Evaluated using adjusted Cox proportional hazards models to estimate VE as $1 - \text{Hazard Ratio (HR)}$ via merged medical claims and state vaccine registries.

7.2 Secondary & Exploratory Endpoints - VE against COVID-19-associated emergency department (ED), urgent care (UC), and outpatient visits. - Financial burden of COVID-19, including total healthcare costs, out-of-pocket expenditures, and payer expenditures for COVID-19 encounters.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Post-marketing safety assessed by screening databases for Adverse Events of Special Interest (AESI) diagnostic codes. **Serious Adverse Events (SAEs):** Identified retrospectively via hospital admissions and mortality data in the HealthVerity database. **Data Monitoring Committee (DMC):** N/A (Retrospective secondary database study).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All individuals meeting the continuous enrollment criteria, further stratified by age (adults vs. pediatrics) and risk factors. **Sample Size Justification:** Descriptive and comparative analyses utilize all available, eligible real-world registry data during the observational window to maximize statistical power. **Handling of Missing Data:** Participants with demographic mismatches across datasets are excluded. Time-dependent Cox proportional hazards models incorporate time-fixed covariates determined from the look-back window.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: Compliant with observational research standards, ensuring complete removal of patient identifiers (HIPAA compliant). **Monitoring Plan:** Monthly refreshes of claims and vaccine registries from HealthVerity. **Audit Trail:** Data extracted using standardized macro scripts. HealthVerity claims, mortality, and vaccine registries are stored within Pfizer's separate secure infrastructure; discrepancies are resolved by independent programmers prior to data locks.

Primary ID: C4591074 **Registry Number:** NCT06923137 **Sponsor/Lead Organization:** Pfizer **Study Title:** A Study to Learn About How Well Yearly Updates to the COVID-19 Vaccine Work to Protect People From COVID-19 and How Much Money People Spend on Healthcare for COVID-19 **Official Regulatory Title:** Effectiveness of BNT162b2 Annual Updated Formulations, Using Administrative Claims Data Tokenized to State Vaccine Registries

Primary Condition: SARS-CoV-2 Infection / COVID-19 **Diagnosis Threshold:** Clinician-assigned diagnostic codes in healthcare claims data. **Medical Methodology:** Post-marketing Observational Database Surveillance **Optimization Target:** Prevention of COVID-19 hospitalization and significant reduction in pandemic-associated economic burden.

Management Pathway: Real-world tracking of US immunization campaigns. **Education Protocol (HCP):** N/A (Retrospective database protocol) **Education Protocol (Patient):** N/A **Quality Audit Mechanism:** Dual-programmer script verification and automated claims data validation constraints.

Total Duration: Follow-up runs continuously from index date to the end of the data availability/study window. **Onsite Visit Cadence:** N/A (Decentralized/Claims-based) **Remote Monitoring Frequency:** Continuous evaluation via monthly database refreshes. **Checklist Review Interval:** N/A

Role	Name	Signature	Date	---	---	---	---
Principal Investigator	Pfizer Medical Lead	_____	[DD-MMM-YYYY]				
	Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			
	Lead Statistician	[Name]	_____	[DD-MMM-YYYY]			